Table 2.2 : Some of common allotropes of p-block elements

Element	Most common allotropes
Boron	Amorphous boron, α-rhombohedral boron, β-rhombohedral boron, γ-orthorhombic boron, α-tetragonal boron, β-tetragonal boron
Carbon	Diamond, Graphite, Graphene, Fullerenes, Carbon nanotubes
Silicon	Amorphous silicon, crystalline silicon
Germanium	α-germanium, β-germanium
Tin	Grey tin, white tin, rhombic tin, sigma tin
Phosphorus	White phosphorus, Red phosphorus, Scarlet phosphorus, Violet phosphorus, Black phosphorus.
Arsenic	Yellow arsenic, grey arsenic & Black arsenic
Anitimony	Blue-white antimony, Yellow, Black
Oxygen	Dioxygen, ozone
Sulphur	Rhombus sulphur, monoclinic sulphur
Selenium	Red selenium, Grey selenium, Black selenium, Monoclinic selenium,
Tellurium	Amorphous & Crystalline

2.2 Group 13 (Boron group) elements:

2.2.1 Occurrence:

The boron occurs mostly as borates and its important ores are borax - $Na_2[B_4O_5(OH)_4].8H_2O$ and kernite - $Na_2[B_4O_5(OH)_4].2H_2O$. Aluminium is the most abundant metal and occurs as oxides and also found in aluminosilicate rocks. Commercially it is extracted from its chief ore, bauxite ($Al_2O_3.2H_2O$). The other elements of this group occur only in trace amounts. The other elements Ga, In and Tl occur as their sulphides.

2.2.2 Physical properties:

Some of the physical properties of the group 13 elements are listed below

Table 2.3 Physical properties of group 13 elements

Property	Boron	Aluminum	Gallium	Indium	Thallium
Physical state at 293 K	Solid	Solid	Solid	Solid	Solid
Atomic Number	5	13	31	49	81
Isotopes	^{11}B	^{27}Al	^{69}Ga	^{115}In	^{205}Tl

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